ACT / ALOHA · arXiv 2304.13705 · RSS 2023

Visionary

11+3

Short generative action trajectories
as motor primitive · chunk+overlap stays

Oct 12 paper club · VLA Architectures

Spine: action chunking as motor contract — TE ≠ receding ≠ open-loop ≠ soft-inpaint

Zhao, Kumar, Levine, Finn · Stanford / Berkeley / Meta

Durable idea

Short generative action trajectories

as the robot’s motor primitive

Churns

CVAE → diffusion
→ flow → hybrids

Stays

chunk + overlap
execution contract

2026 bet

matched commit_s
+ stages + Hz-honest
teleop

What to keep vs drop

KEEP (non-negotiable)

- Action chunking
- Overlap / boundary consistency
- High-rate action interface
- Bimanual coordination data
- commit_seconds as knob

OPTIONAL / REPLACE

- CVAE head → FM / DiT
- Classical TE → RTC/receding
- Per-task 80M → VLA backbone
- z=0 deterministic decode
- k=100@50Hz cargo-cult

Chunking non-negotiable · CVAE+TE optional · prefer RTC/receding over naïve open-loop

Skip Discord / Drive digressions

2026 BEHAVIOR bets · no Discord / Drive

1	Start OpenPI $\pi_{0.5}$ / GR00T N1.7 — not from-scratch ACT
2	Port horizon / overlap / correlated noise — not CVAE
3	Prefer RTC / receding over naïve full open-loop
4	Chunk motors + chunk semantics (stages / System 2)
5	Hz-honest teleop for dexterous segments
6	Cite ACT as control prior, not drop-in household baseline

Control schemes — do NOT conflate

Chunking is shared · TE ≠ receding ≠ open-loop ≠ soft-inpaint

Scheme	Mechanism	Commit / loop
ACT TE	Every-step query; exp-avg overlapping preds for same t	ensembled 1-step @ 50 Hz
DP receding	Predict Tp, exec Ta < Tp, replan	Ta≈8 @ ~10 Hz
π_0 open-loop	H=50; exec prefix; NO TE (hurt early for them)	open-loop prefix @ 20–50 Hz
RTC soft-inpaint	Async; freeze prefix; soft-mask remainder	latency-aware async
BEHAVIOR'25	Predict 30 / exec 26 / keep 4 + correlated FM	receding ~30 Hz

Map: k/50 ↔ DP Ta·Δt ↔ π_0/Comet H/Hz · Defaults: k=100 (~2s @ 50 Hz); L1+βKL; test z=0

Club line

ACT coined chunk + ensemble for 50 Hz fine bimanual BC

Later VLAs keep chunks, often drop ensemble,

retune H / commit Hz · TE ≠ receding ≠ open-loop ≠ soft-inpaint

ACT+TE → **DP recede** → **π_0 open** → **RTC inpaint** → **BEH 30/26/4**

Map horizons by commit_seconds — not raw k or H

Genealogy · CLEAR lineage

ALOHA+ACT (CVAE+TE) ‖ DP → Mobile/ALOHA2 → $\pi_0/\pi_{0.5}$ → RTC → BEHAVIOR’25/’26

2023 ACT

CVAE chunks
+ temporal ensemble
k=100 @ 50 Hz

2023 DP ‖

diffusion chunks
+ receding Ta<Tp
parallel ancestor

2024

Mobile ALOHA
ALOHA 2
data engines

2024–25 π_0

FM chunks H=50
NO TE (hurt early)
cites ACT

2025 RTC

soft-inpaint
async freeze prefix
latency-aware

’25/’26

BEHAVIOR
30/26/keep4
+ corr. FM

STAYED: generative continuous chunks · overlap / consistency at boundaries

MOVED: CVAE→diffusion→flow · TE→receding→soft-inpaint · tabletop→mobile→household VLA

Horizon map · normalize by commit seconds

ACT	k=100 @ 50 Hz	**~2.0 s chunk**	TE → 1-step ensembled
DP	Tp=16, Ta=8 @ 10 Hz	**~0.8 s commit**	receding replan
π_0	H=50 @ 20–50 Hz	**~1.0–2.5 s**	open-loop prefix
BEHAVIOR’25	30 / exec 26 @ ~30 Hz	**~0.87 s**	keep 4 + inpaint

k/50 ↔ Ta·Δt ↔ H/Hz — Empiricist + Archaeologist shared language

Action chunking vs Temporal Ensembling

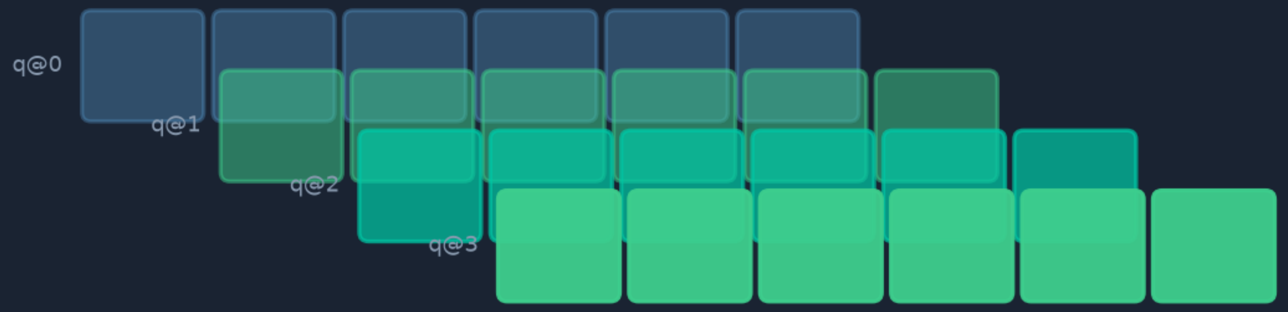
Naïve open-loop every k steps vs every-step query + exp-weighted overlap

Naïve chunking (open-loop)

observe → predict k → execute all k → replan · jerky at boundaries

ACT Temporal Ensembling

Query every tick · buffer overlapping preds for same absolute t · $a_t = \Sigma\, w_i\, \hat{A}_t[i] \,/\, \Sigma\, w_i$ · $w_i = \exp(-m \cdot i)$



Same-t fusion

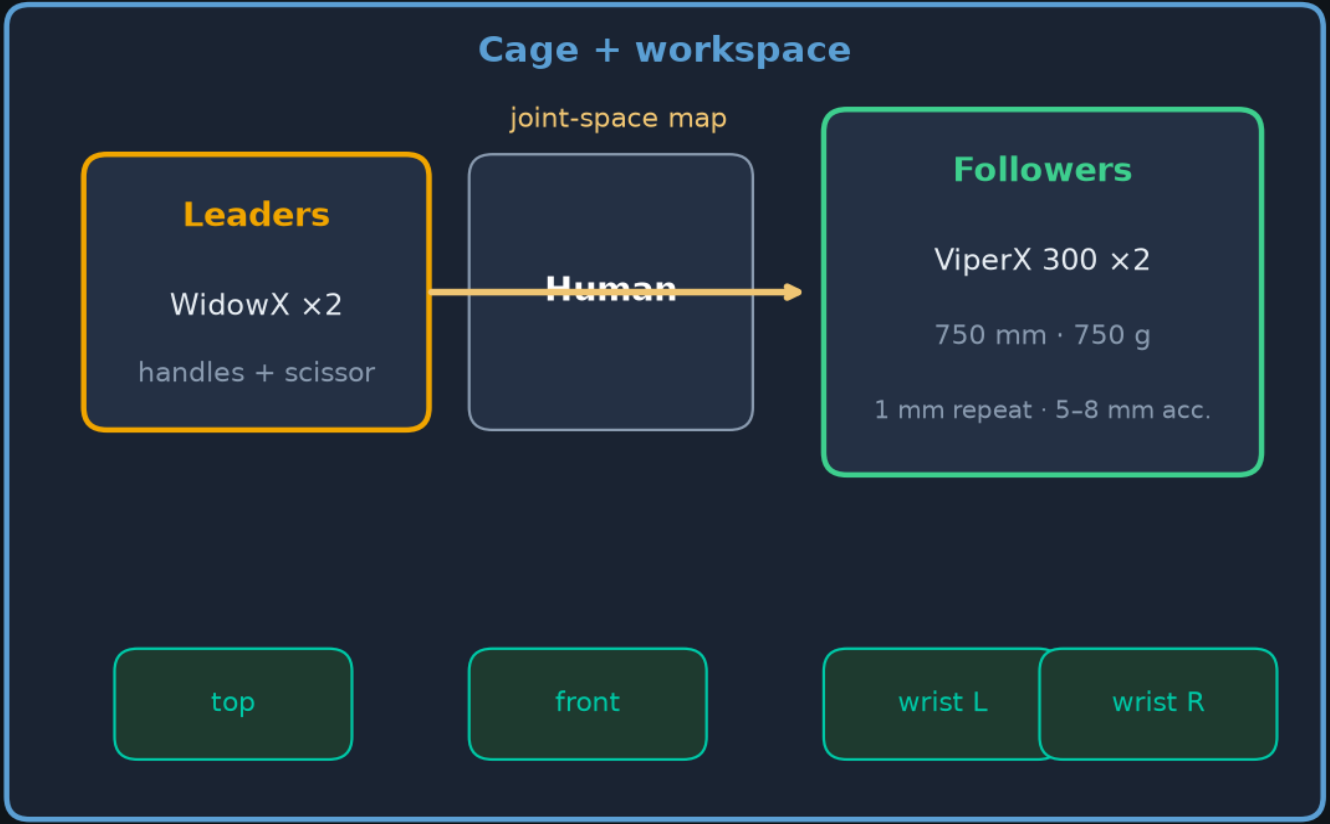
$w = [e^{0}, e^{-m}, e^{-2m}, \ldots]$

recent obs → faster (small m)

+~3.3 pp ACT (paper)

ALOHA hardware · <$20k leader–follower bimanual

WidowX leaders → ViperX followers · joint-space teleop · 4 RGB · 50 Hz



Budget

<$20k total
2×ViperX + 2×WidowX
+ cage + cams

Control

50 Hz record/control
Absolute joint targets
Dynamixel PID

Design

Low-cost · versatile
Repairable · <2 h build
Rubber-band gravity assist

Follow-ons: Mobile ALOHA (2401.02117) · ALOHA 2 (2405.02292) — data engines, not just demos

Durable idea: short generative action trajectories

Generative family churns (CVAE→diff→FM)
chunk + overlap execution stays · no Discord/Drive

Q&A · keep the spine clear

Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware

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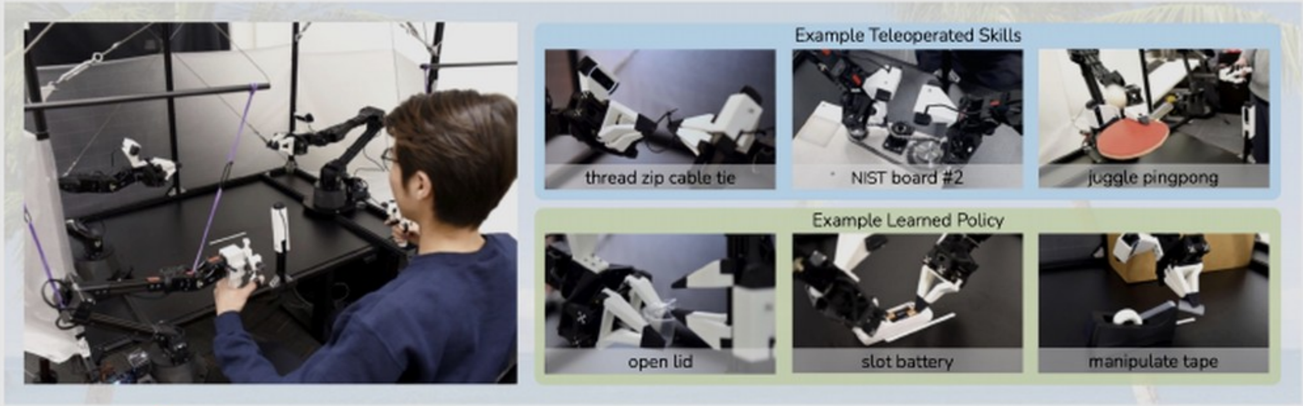

Fig. 1: *ALOHA : A Low-cost Open-source Hardware System for Bimanual Teleoperation*. The whole system costs <$20k with off-the-shelf robots and 3D printed components. *Left:* The user teleoperates by backdriving the leader robots, with the follower robots mirroring the motion. *Right: ALOHA* is capable of precise, contact-rich, and dynamic tasks. We show examples of both teleoperated and learned skills.

***Abstract*—Fine manipulation tasks, such as threading cable ties or slotting a battery, are notoriously difficult for robots because they require precision, careful coordination of contact forces, and closed-loop visual feedback. Performing these tasks typically requires high-end robots, accurate sensors, or careful calibration, which can be expensive and difficult to set up. *Can learning enable low-cost and imprecise hardware to perform these fine manipulation tasks?* We present a low-cost system that performs end-to-end imitation learning directly from real demonstrations, collected with a custom teleoperation interface. Imitation learning, however, presents its own challenges, particularly in high-precision domains: errors in the policy can compound over time, and human demonstrations can be non-stationary. To address these challenges, we develop a simple yet novel algorithm, *Action Chunking with Transformers (ACT)*, which learns a generative model over action sequences. ACT allows the robot to learn 6 difficult tasks in the real world, such as opening a translucent condiment cup and slotting a battery with 80-90% success, with only 10 minutes worth of demonstrations. Project website: tonyzhaozh.github.io/aloha**

off the table. Next, one of the right fingers approaches the cup from below and pries the lid open. Each of these steps requires high precision, delicate hand-eye coordination, and rich contact. Millimeters of error would lead to task failure.

Existing systems for fine manipulation use expensive robots and high-end sensors for precise state estimation [29, 60, 32, 41]. In this work, we seek to develop a low-cost system for fine manipulation that is, in contrast, accessible and reproducible. However, low-cost hardware is inevitably less precise than high-end platforms, making the sensing and planning challenge more pronounced. One promising direction to resolve this is to incorporate learning into the system. Humans also do not have industrial-grade proprioception [71], and yet we are able to perform delicate tasks by learning from closed-loop visual feedback and actively compensating for errors. In our system, we therefore train an end-to-end policy that directly maps RGB images from commodity web cameras to the actions.